Introduction

In today's complex, fast-moving technology environments, engineering teams are under immense pressure. Systems are more distributed, data volumes are growing exponentially, and incidents are becoming more frequent and harder to manage. To navigate this complexity, teams need a strong observability foundation.

While visibility—real-time insight across infrastructure, applications, and services—is foundational, true observability goes beyond just monitoring. It's about taking swift, confident action on observability insights to reduce resolution times and improve customer experiences.

So how do you move from visibility to action?

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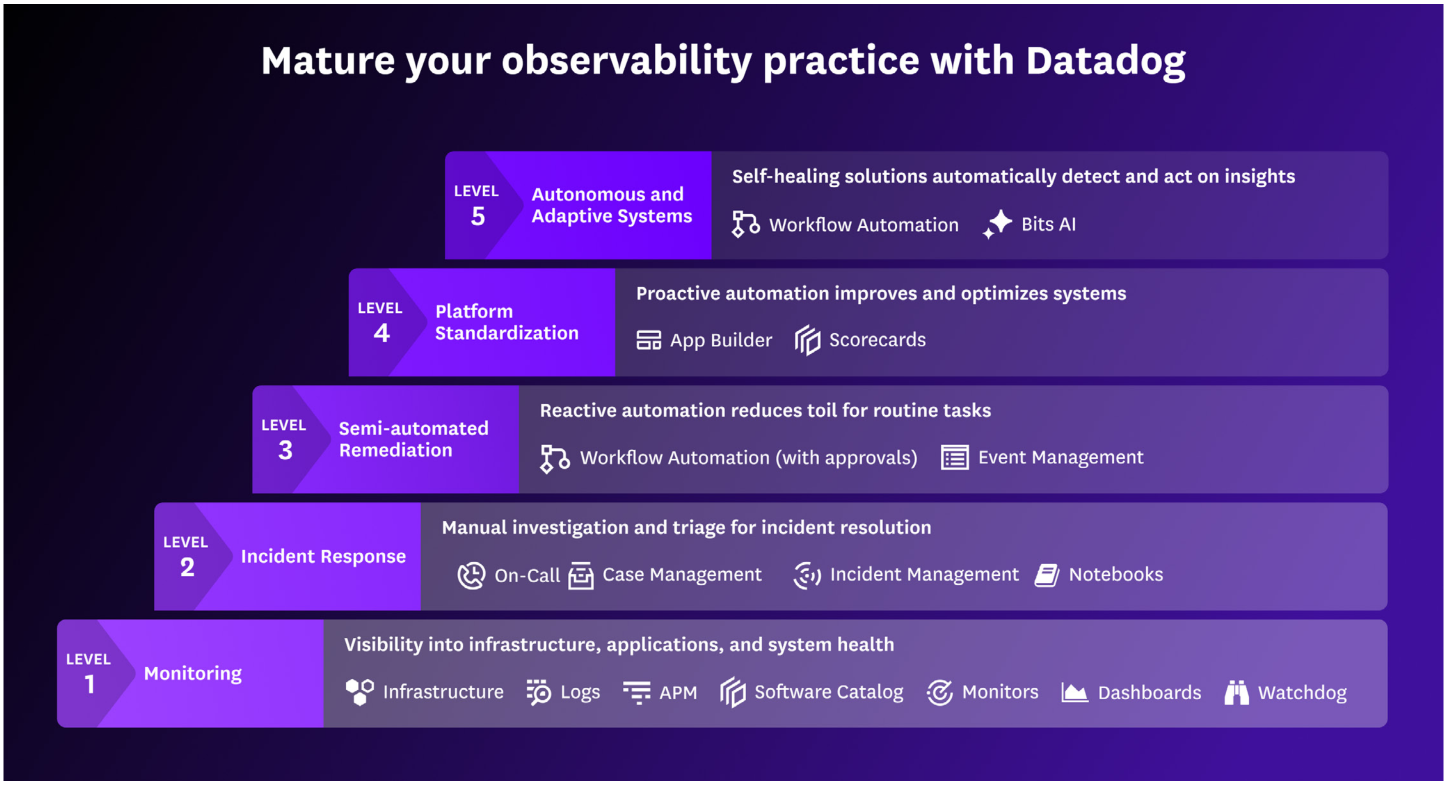
From Insights to Action: Introducing the Maturity Framework

When we say you can't just have visibility—you also need to take action—we're talking about the difference between simply knowing there's a problem and being equipped to resolve it quickly, intelligently, and with minimal manual effort. That's where maturity comes in.

This eBook introduces the Observability Maturity Framework, outlining a path from basic monitoring to adaptive, self-healing systems. This framework details how teams can evolve their practice to not only detect problems but fix them faster, optimize systems, and build resilience at scale. Levels 2 through 5 of this framework represent a shift: from visibility to action.



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Laying the Groundwork for Reliable Response During Incidents

Monitoring is the foundation of any observability practice. It gives you the visibility needed to start understanding your systems and their behavior. But without a clear and consistent way to respond, teams risk falling into reactive patterns—scrambling to coordinate during incidents, duplicating effort, or wasting time hunting for the right context. That's why the first step beyond monitoring isn't automation or AI—it's **structured incident response**.

WITHOUT STRUCTURED INCIDENT RESPONSE, TEAMS STRUGGLE WITH:

- A flood of metrics, logs, and traces without clear prioritization
- Remediation steps stored in stale docs, personal notes, or tribal Slack threads
- Unclear ownership, leading to delays, confusion, or duplicated efforts

WITH STRUCTURED INCIDENT RESPONSE, TEAMS GAIN:

- A well-defined incident lifecycle from detection to triage to postmortem
- Runbooks directly linked to incidents, providing critical context in real time
- Live updates across Slack and Datadog's UI to streamline collaboration

Datadog enables this shift with integrated tools like On-Call and Incident Management.

On-Call is our newly launched, fully integrated paging solution. Unlike standalone paging tools, it's built natively into Datadog, so there's no need for extra integrations, and there's no context loss. Responders can receive, acknowledge, and escalate alerts directly within Datadog—streamlining the response and reducing friction.

> "With Datadog On-Call we now have integrated observability, paging and incident response in one platform that helps us get the right person involved with a page as fast as possible to triage product stability."
>
> **Matthew Green**
> Staff Engineer at Torc Robotics

Once an incident is declared, **Incident Management** provides immediate access to the relevant telemetry, timelines, and collaboration tools needed to drive resolution quickly. And to keep everyone in the loop, our new Status Pages feature automatically shares real-time incident updates with both internal and external stakeholders.

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Accelerating Response with Automation

But, even with clear processes in place, responders often spend valuable time executing the same manual steps to apply fixes, update stakeholders, or dig through documentation. These tasks are predictable, time-consuming, and prone to human error. That's why the natural next step in observability maturity is **automation**. By automating what can, and should, be automated, teams can reduce toil, speed up resolution, and stay focused on solving the incidents that truly require human insight.

> "Reacting quickly to changes in complex systems requires automation. Datadog provides an intuitive way to leverage context-rich data to automate monitor creation, update integrations, and reduce complexity and overhead in reliability and incident response engineering."
>
> **Ryan Kleeberger**
> SRE at Protolabs

WITHOUT AUTOMATION, TEAMS STRUGGLE WITH:

- Responders copy and paste scripts, dig through logs, or manually notify teammates during each incident
- Dozens of redundant alerts trigger for the same issue, overwhelming teams without pointing to root cause
- Engineers firefighting the same recurring issues instead of solving underlying problems

WITH AUTOMATION, TEAMS GAIN:

- One-click automations for common fixes like restarting pods or scaling services without leaving the incident UI
- Machine learning-powered correlation that intelligently groups related alerts into cases to reduce noise and reveal patterns
- Repeatable workflows that are built once and reused across services for consistent, scalable incident resolution

Datadog empowers this shift with tightly integrated automation tools.

Event Management cuts through alert noise by intelligently correlating related signals into a single, actionable event. These consolidated events provide rich context - such as impacted services, related dashboards, runbooks, and historical patterns - so responders can assess severity and act immediately.

Workflow Automation enables teams to encode response playbooks into reusable workflows that can be triggered by alerts, schedules, or manual input. Whether it's restarting a failed Kubernetes pod, blocking a suspicious IP, or scaling infrastructure, automation handles the repetitive work while built-in approvals ensure human oversight where needed.

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Turning Automation into Scalable Best Practices

Automation accelerates incident response—but without alignment, it can create silos. When teams build workflows independently, they risk duplicating effort, introducing inconsistency, and making coordination harder as the organization grows.

To truly operationalize automation at scale, teams need a shared foundation—one that codifies best practices, enforces consistency, and empowers everyone to move faster together. This is the next step in observability maturity: **standardization**.

At this stage, the focus shifts to creating shared, reusable standards that don't just promote best practices—they embed them directly into the way teams operate.

WITHOUT STANDARDIZATION, TEAMS STRUGGLE WITH:

- Every team builds, tests, and deploys differently, slowing velocity and increasing risk
- No consistent SLOs or dashboards to track performance or reliability across teams
- Teams operate in silos with no common frameworks or expectations for quality

WITH STANDARDIZATION, TEAMS GAIN:

- Predefined templates and golden paths that make it easy to ship consistent, compliant services
- Standard SLOs and observability scorecards give org-wide visibility into service readiness
- Embedded guardrails that align teams to org-wide best practices ensure security, performance, and reliability

Datadog helps drive this shift with tools like Scorecards and App Builder.

Scorecards continuously assess service maturity across key dimensions like observability coverage, runbook availability, and SLO adherence. They provide a live, objective snapshot of each team's health and performance.

App Builder streamlines delivery with self-service actions, enabling developers to easily scaffold new projects in GitHub, manage Kubernetes deployments or SQS queues, and provision resources like EKS clusters and RDS instances - all while automatically adhering to internal standards for security, performance, and compliance.

> "App Builder empowers our global workforce to take action. What was once a team-specific ticket can now be handled independently through self-service, eliminating the need for another tool and extra training."
>
> **Lukas Deutz**
> Systems Engineer, NationBuilder

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Enabling Intelligent, Autonomous Operations

Once standards are embedded and best practices are scaled across teams, the next frontier is removing manual effort altogether. Even with golden paths, human intervention can still slow down response, introduce variability, or delay resolution.

That's why the final stage of observability maturity is all about autonomous operations - empowering systems to take action on their own using predefined logic, rich context, and AI-driven intelligence. At this stage, AI and automation come together to create a closed-loop response system that minimizes human intervention and maximizes uptime.

WITHOUT AUTONOMY, TEAMS FACE:

- Engineers comb through dashboards and metrics to pinpoint root cause, wasting valuable minutes
- Teams must coordinate handoffs, approvals, and actions manually across tools and functions
- Remediation depends on tribal knowledge instead of system-driven insights or automation

WITH AUTONOMY, TEAMS GAIN:

- Intelligent triage highlights likely root causes and impacted systems instantly, reducing the burden on engineers
- AI proposes context-aware fixes based on past incidents and system behavior, accelerating remediation
- Common issues are handled without human intervention, enabling systems to self-heal in real time

Datadog powers this transformation with Workflow Automation and Bits AI.

With fully-autonomous **Workflow Automation**, teams can build workflows that operate without human involvement. These workflows can be triggered automatically by monitors or security signals, enabling systems to take immediate action—such as restarting a service, scaling infrastructure, or opening a ticket—without waiting for a person to step in.

This vision becomes reality with **Bits AI**, a suite of intelligent AI capabilities that work across your critical monitoring, development, and security workflows. Bits can autonomously investigate alerts, suggest code fixes, review security signals, and take action on external systems with full contextual, organizational, and operational awareness.

> "Bits AI is helping our team save time triaging bugs, accelerate fix generation, and respond to incidents more efficiently, ultimately enabling us to resolve issues faster and stay focused on delivering value."
>
> **Manan Kothari**
> Staff Site Reliability Engineer, project44

Taking the Next Step

Reaching Level 5 doesn't happen overnight. And that's okay. Most teams move through these levels gradually, and many are at multiple stages at once, depending on the team or service.

The key is to start with where you are, assess your current pain points, and look for quick wins that create momentum.

Ask yourself

- Are incidents still being managed manually?
- Are engineers overwhelmed by alerts?
- Are best practices being followed or reinvented on every team?

If the answer to any of these is yes, you don't need a complete overhaul. Tools like Workflow Automation, Scorecards, and App Builder are easy to adopt one team at a time and can immediately reduce friction without requiring full platform transformation.

Conclusion

Observability maturity isn't about checking boxes; it's about building operational resilience that scales with your team, your systems, and your ambitions. Whether you're just getting started with monitoring or ready to explore autonomous remediation, Datadog can help you turn insights into action.

To learn more about how to mature your observability practice, reach out to your Datadog Customer Success Manager or email actions-all@datadoghq.com to get in touch with our team.